

IBM DATA SCIENCE PROFESSIONAL CERTIFICATE • APPLIED CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

A data science capstone project: collecting, wrangling, exploring, and modeling Falcon 9 launch data to predict whether the first stage will land successfully.

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GitHub: <https://github.com/deepak1145460-design/Data-science-capstone>

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What this project set out to do — and what we found



Methods

Collected launch data via the SpaceX REST API and by scraping Wikipedia's Falcon 9 launch-records table, then wrangled both into a single analysis-ready dataset with an engineered landing-success label.



Analysis

Explored the data with visual EDA (Matplotlib/Seaborn) and SQL, then built an interactive Folium map and a Plotly Dash app to analyze launch sites, payloads, and outcomes.



Modeling

Trained and tuned four classifiers (Logistic Regression, SVM, Decision Tree, KNN) with GridSearchCV to predict first-stage landing success from flight parameters.



Key Result

Landing success rate rose from near 0% in 2010–2013 to over 80% by 2019–2020. The tuned Logistic Regression / SVM models reached 83.3% test accuracy, the best of the four candidates.

Background and problem statement

Why this matters

- SpaceX advertises Falcon 9 launches at \$62M, far below competitors' \$165M+, largely because SpaceX reuses the first stage instead of discarding it.
- If we can predict whether a launch's first stage will land successfully — before it flies — we can estimate the true cost of a launch, which is valuable for a company that wants to compete with SpaceX for a contract.
- Problem statement: using historical launch data (site, payload, orbit, booster version, flight number), predict the binary outcome — did the first stage land successfully or not?

Project Questions

- Which variables most affect landing outcome?
- How has success rate changed over time?
- Which orbit types and launch sites are riskiest?
- Which classification model best predicts a landing?

Notebook 01: pulling Falcon 9 launch data from api.spacexdata.com

1

GET requests

Call 4 REST endpoints — launches, rockets, payloads, launchpads, cores — to pull raw JSON.

2

Resolve IDs

Each launch record only stores IDs; helper functions look up booster, launchpad, payload and core names.

3

Flatten & assemble

Merge the resolved fields into one flat DataFrame — one row per launch.

4

Filter & clean

Keep Falcon 9 only (drop Falcon 1), renumber flights, fill missing payload mass with the column mean.

Result

90 Falcon 9 launch records

17 raw fields → flattened, human-readable

Saved as the base dataset for wrangling & modeling

GitHub URL for this notebook

```
https://github.com/deepak1145460-design/Data-science-capstone/blob/main/01\_data\_collection\_api\_.ipynb
```

Notebook 02: scraping Wikipedia's Falcon 9 & Falcon Heavy launch table

1

Fetch HTML

Request the Wikipedia page "List of Falcon 9 and Falcon Heavy launches" with requests.

2

Locate tables

Use BeautifulSoup to find the launch-record <table> elements on the page.

3

Parse rows

Programmatically read table headers, then iterate every <tr>/<td> to extract each launch's fields.

4

Build & export

Assemble the parsed rows into a DataFrame — an independent, cross-check dataset.

Purpose

Provides an independent dataset to cross-check
the API-sourced records for consistency
before the two are combined during wrangling

GitHub URL for this notebook

```
https://github.com/deepak1145460-design/Data-science-capstone/blob/main/02\_data\_collection\_webscraping\_.ipynb
```

Notebook 03: cleaning the data and engineering the landing-success label

1

Audit missing values

Reviewed nulls column by column — concentrated in LandingPad, expected for expendable or failed landings.

2

Review distributions

Checked launch-site and orbit-type frequency counts to understand class balance.

3

Encode outcomes

Landing outcome strings (e.g. 'True ASDS', 'False Ocean', 'None None') mapped to a bad/good outcome set.

4

Engineer Class label

Class = 1 only when landing_success is True; every other outcome (failure, no attempt) = 0.

**Overall landing success rate in the wrangled dataset: 66.67%
(90 launches, 18 fields incl. Class)**

GitHub URL for this notebook

GitHub:
https://github.com/deepak1145460-design/Data-science-capstone/blob/main/03_data_wrangling.ipynb

Notebook 04: charts used and what each one shows

Scatter — Flight Number vs Launch Site

Whether landing success improves as flight number (experience) increases, per site.

Scatter — Payload Mass vs Launch Site

Whether heavier payloads at a given site are linked to lower success.

Bar — Success Rate by Orbit

Ranks orbit types by landing success rate to flag the riskiest orbits.

Scatter — Flight Number vs Orbit

How success by orbit type has evolved as SpaceX gained more flights.

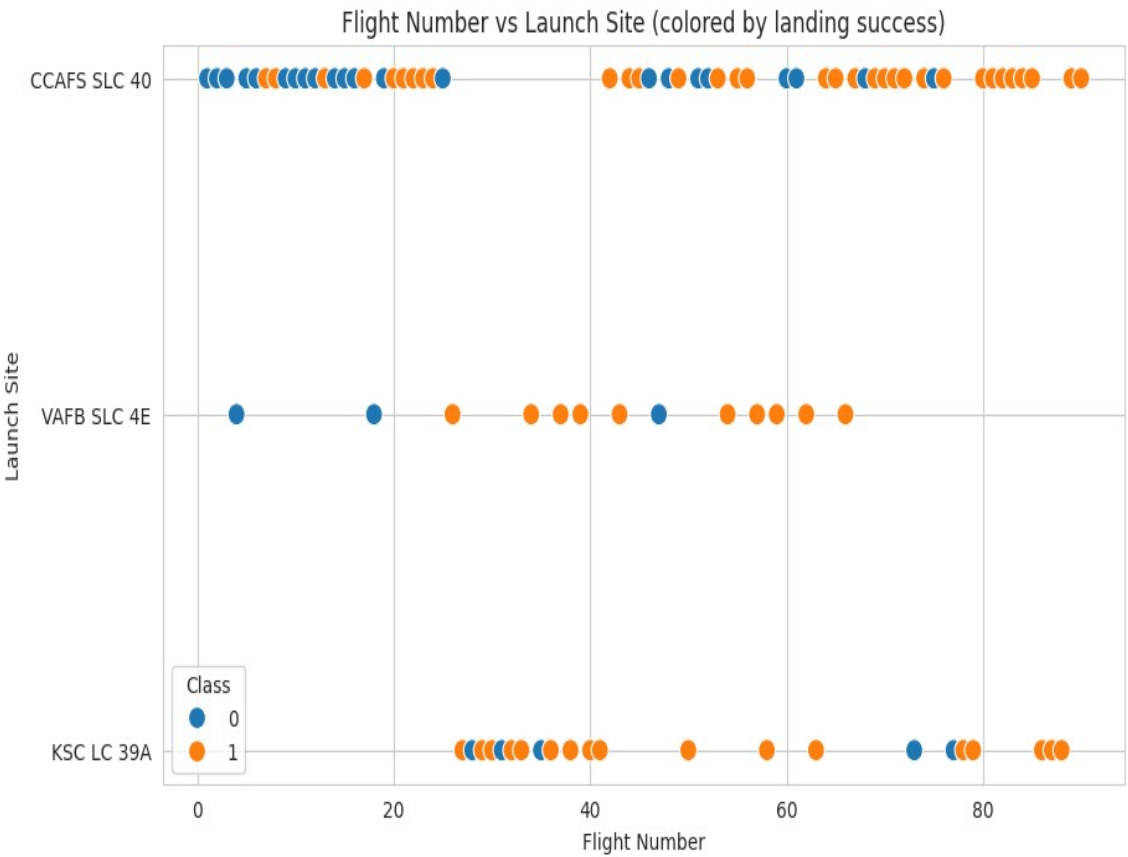
Scatter — Payload Mass vs Orbit

Whether payload mass interacts with orbit type to affect landing odds.

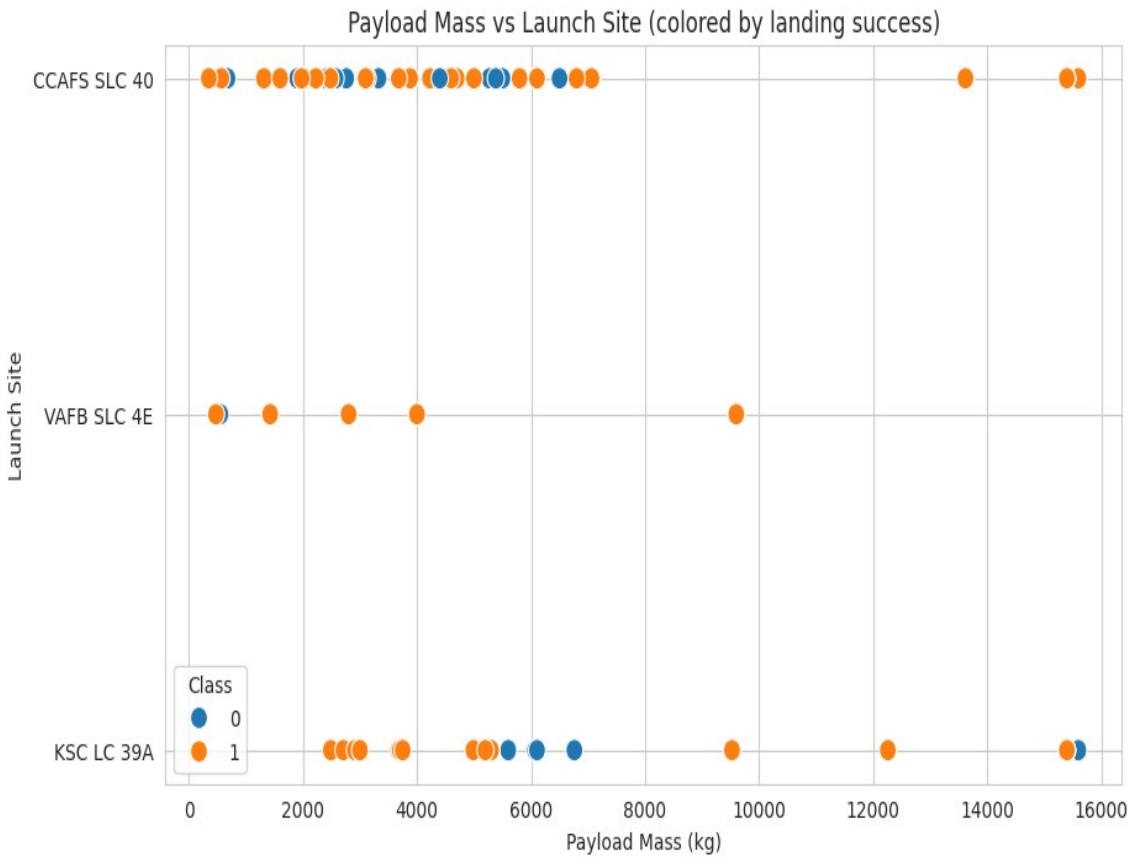
Line — Yearly Success-Rate Trend

The overall trajectory of landing success from 2010 to 2020.

Flight number & payload mass vs. launch site

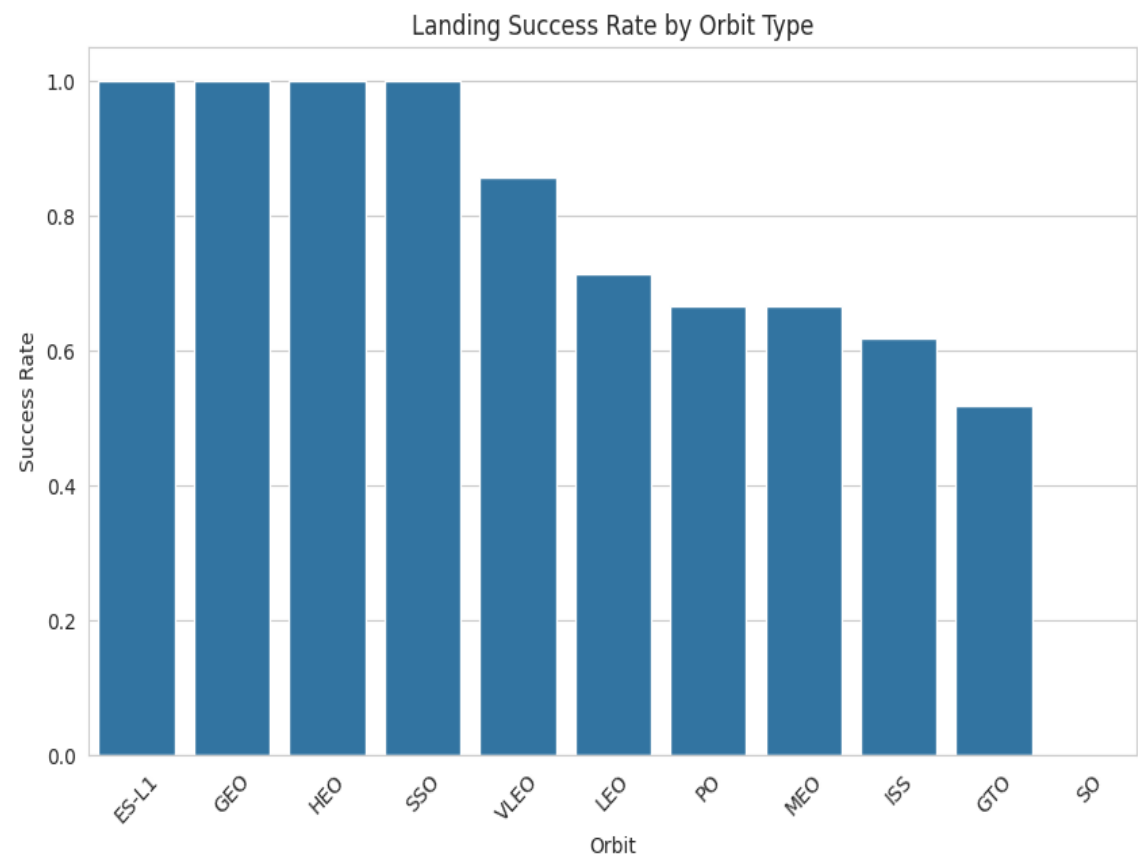


Later flights (higher FlightNumber) skew orange (success) at every site — SpaceX's recovery process visibly matured with experience.

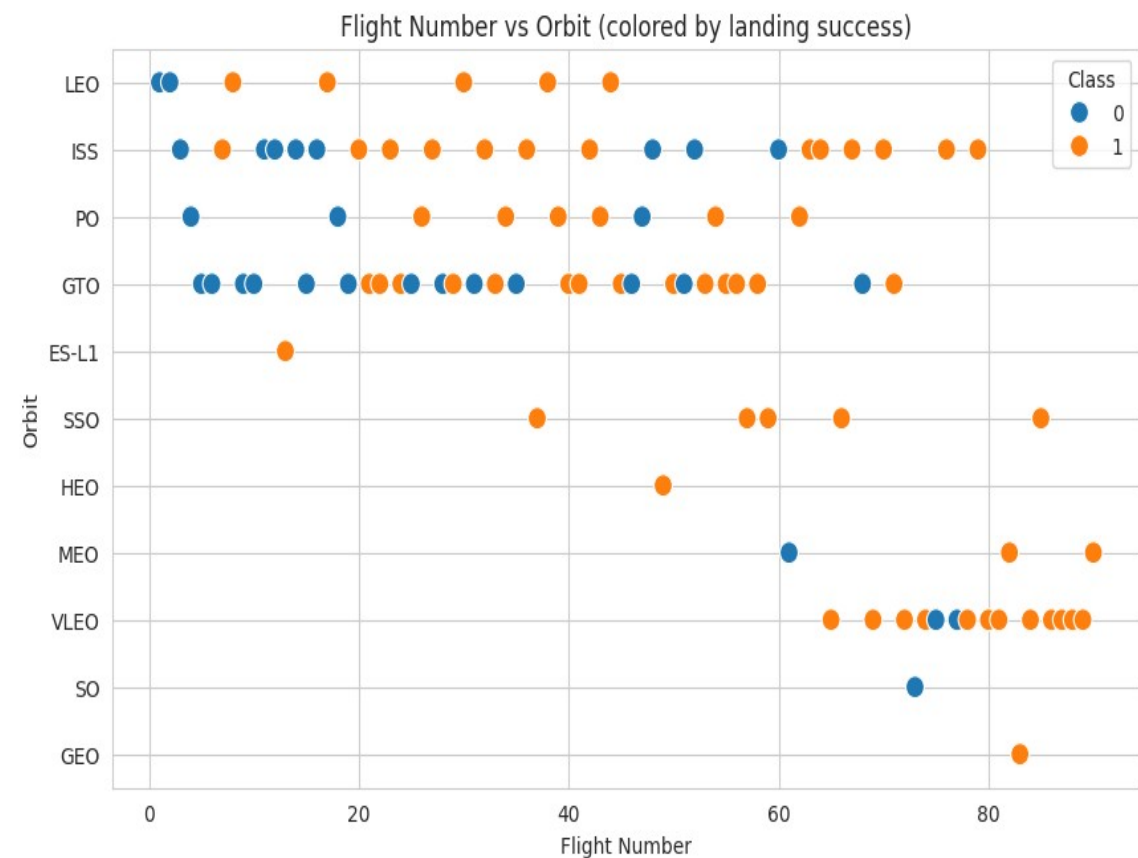


Heavier payloads succeed less consistently, especially at VAFB SLC-4E — payload mass is a meaningful risk factor.

Success rate by orbit & flight number vs. orbit

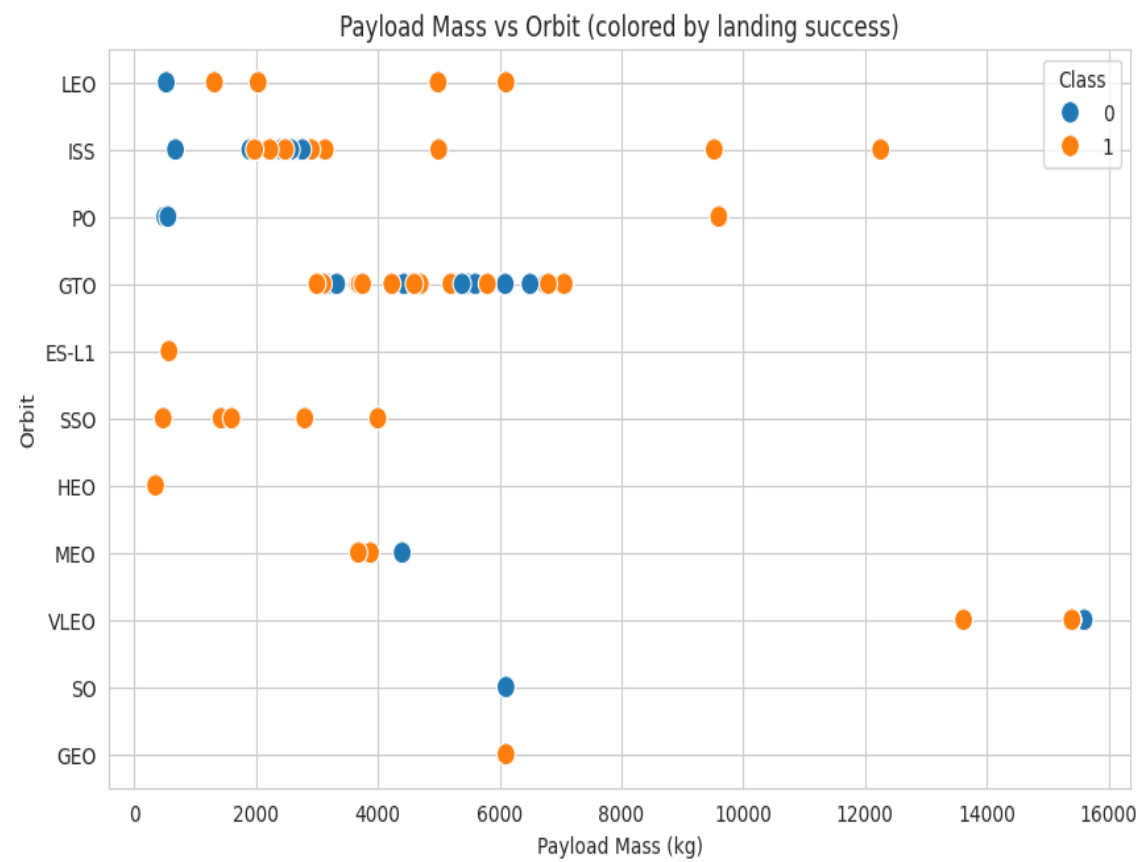


ES-L1, GEO, HEO and SSO orbits show a 100% landing success rate in this dataset; GTO and SO are the riskiest.

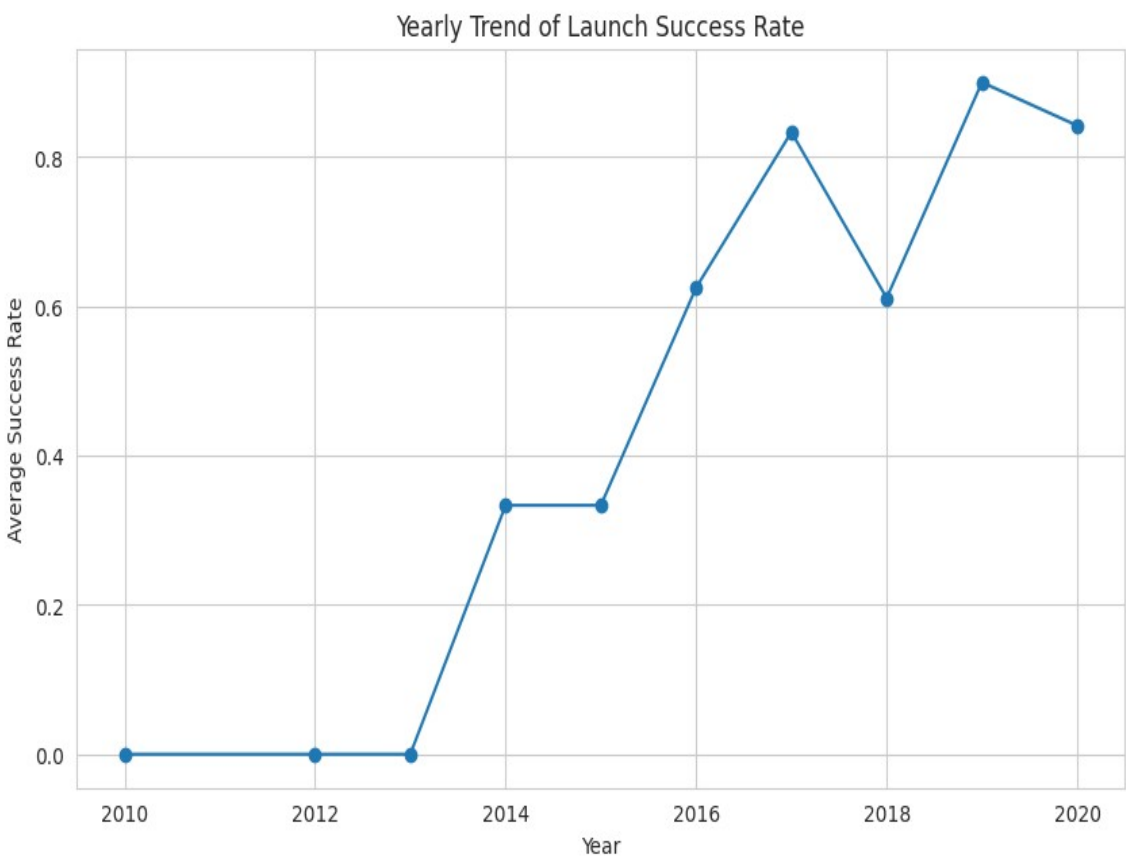


Success in higher-energy orbits (GTO, ISS) also improves at higher flight numbers, tracking overall program maturity.

Payload mass vs. orbit & the yearly success-rate trend



Payload/orbit combinations cluster — very heavy payloads mostly target LEO/ISS, where recovery is comparatively easier.



Success rate was ~0% through 2013, climbed sharply from 2014, and has stayed above 60-90% since 2016.

Notebook 05: 10 queries run against a local SQLite table (SPACEXTBL)

1. Distinct launch sites

2. Sites beginning with 'CCA'

3. Total NASA (CRS) payload mass

4. Average payload mass by booster version

5. Date of first successful ground-pad landing

6. Boosters: drone-ship success, 4000–6000 kg payload

7. Total successes vs. failures

8. Booster versions carrying max payload

9. Launch-site / outcome ranking by year

10. Failure-outcome ranking, 2010-06-04 to 2017-03-20

GitHub URL for this notebook

```
https://github.com/deepak1145460-design/Data-science-capstone/blob/main/05\_eda\_sql\_.ipynb
```

Query outputs: launch sites, payloads, success rates, rankings, time analysis

EDA with SQL — Query Results (05_eda_sql_.ipynb, SPACEXTBL, 90 rows)

1. Distinct Launch Sites

LaunchSite
CCAFS SLC 40
VAFB SLC 4E
KSC LC 39A

2. Sites starting 'CCA' (sample)

FlightNumber	Date	Orbit
1	2010-06-04	LEO
2	2012-05-22	LEO
3	2013-03-01	ISS
5	2013-12-03	GTO
6	2014-01-06	GTO

3. Total NASA/CRS Payload Mass

Total_Payload_Mass_kg
305151.43

4. Avg Payload Mass (Falcon 9)

BoosterVersion	Avg_Payload_kg
Falcon 9	6104.96

5. First Successful Landing

First_Successful_Landing
2014-04-18

6. Boosters (drone-ship, 4-6t)

BoosterVersion	Count
Falcon 9	9.0
Falcon 9	nan
Falcon 9	nan
Falcon 9	nan
Falcon 9	nan
Falcon 9	nan
Falcon 9	nan
Falcon 9	nan
Falcon 9	nan

7. Total Successes vs Failures

Class	Count	Label
0	30	Failure
1	60	Success

8. Booster(s), Max Payload

BoosterVersion	PayloadMass_kg
Falcon 9	15600.0
Falcon 9	15600.0
Falcon 9	15600.0

9. Site/Year Success Rate (sample)

LaunchSite	Year	Success_Rate
CCAFS SLC 40	2017	1.0
CCAFS SLC 40	2019	0.86
KSC LC 39A	2018	1.0
KSC LC 39A	2020	0.71
VAFB SLC 4E	2019	1.0

10. Failure Outcomes (2010-2017)

Outcome	Occurrences
None None	9
True ASDS	5
False ASDS	4
True RTLS	3
True Ocean	3
None ASDS	2
False Ocean	2



Notebook 06 (Folium) & 07_spacex_dash_app.py (Plotly Dash)



Folium Map

- Marks every launch site on an interactive map
- Colors each launch record green (success) / red (failure)
- Adds a mouse-position readout for distance measurement
- Proximity analysis to coastline, railway, highway & nearest city

GitHub URL

```
GitHub: https://github.com/deepak1145460-design/Data-science-capstone/blob/main/06\_folium\_launch\_site\_analysis\_.ipynb
```



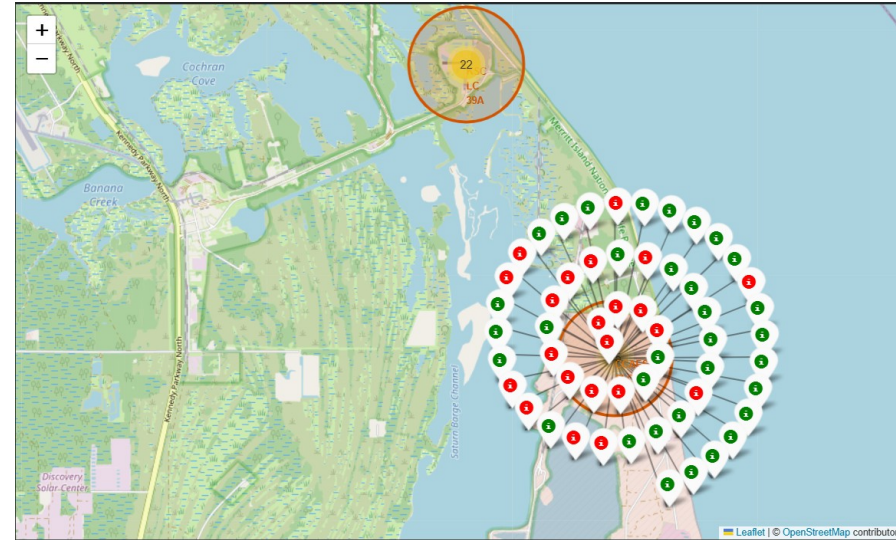
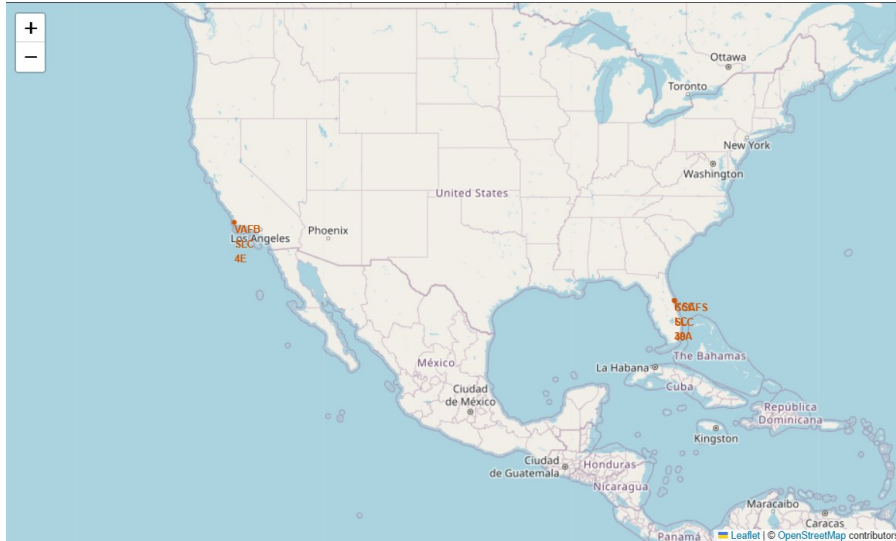
Plotly Dash App

- Dropdown to pick a launch site (or "All Sites")
- Pie chart: success-count share by site
- Range slider to filter by payload mass
- Scatter: payload mass vs. success, colored by booster version

GitHub URL

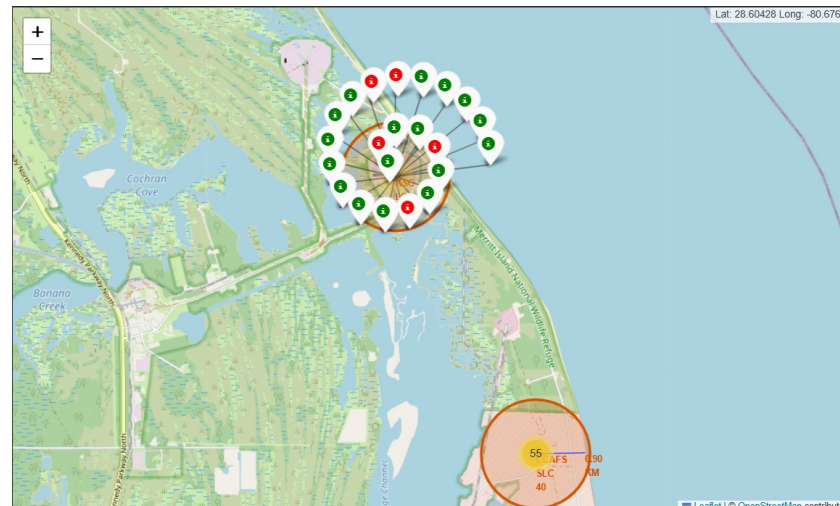
```
GitHub: https://github.com/deepak1145460-design/Data-science-capstone/blob/main/07\_spacex\_dash\_app\_.py
```

Launch site markers, launch outcomes, and proximity analysis

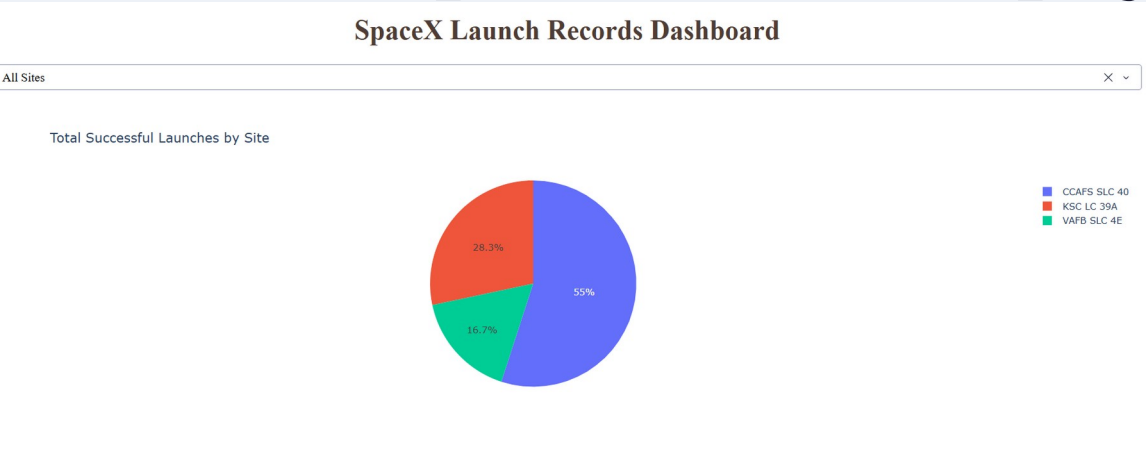


Markers are colored green for a successful first-stage landing and red for a failure, plotted across all three launch sites — CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.

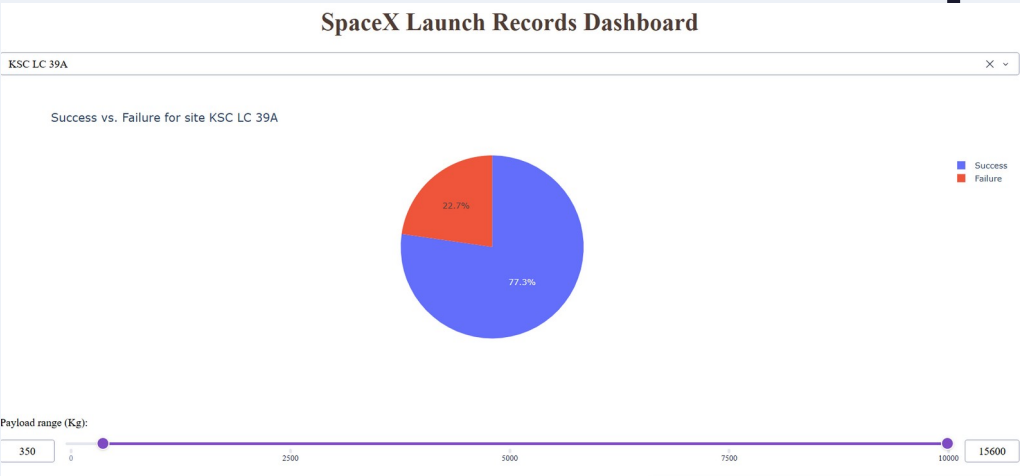
Proximity analysis confirms each pad sits close to the coastline for overwater-abort safety, and well clear of highways, railways, and the nearest city — consistent with FAA siting rules for orbital launch facilities.



Success pie chart and payload-vs-outcome scatter plot



CCAFS SLC-40 contributes the largest share of successful launches, followed by KSC LC-39A and VAFB SLC-4E, reflecting each site's launch volume in the dataset.

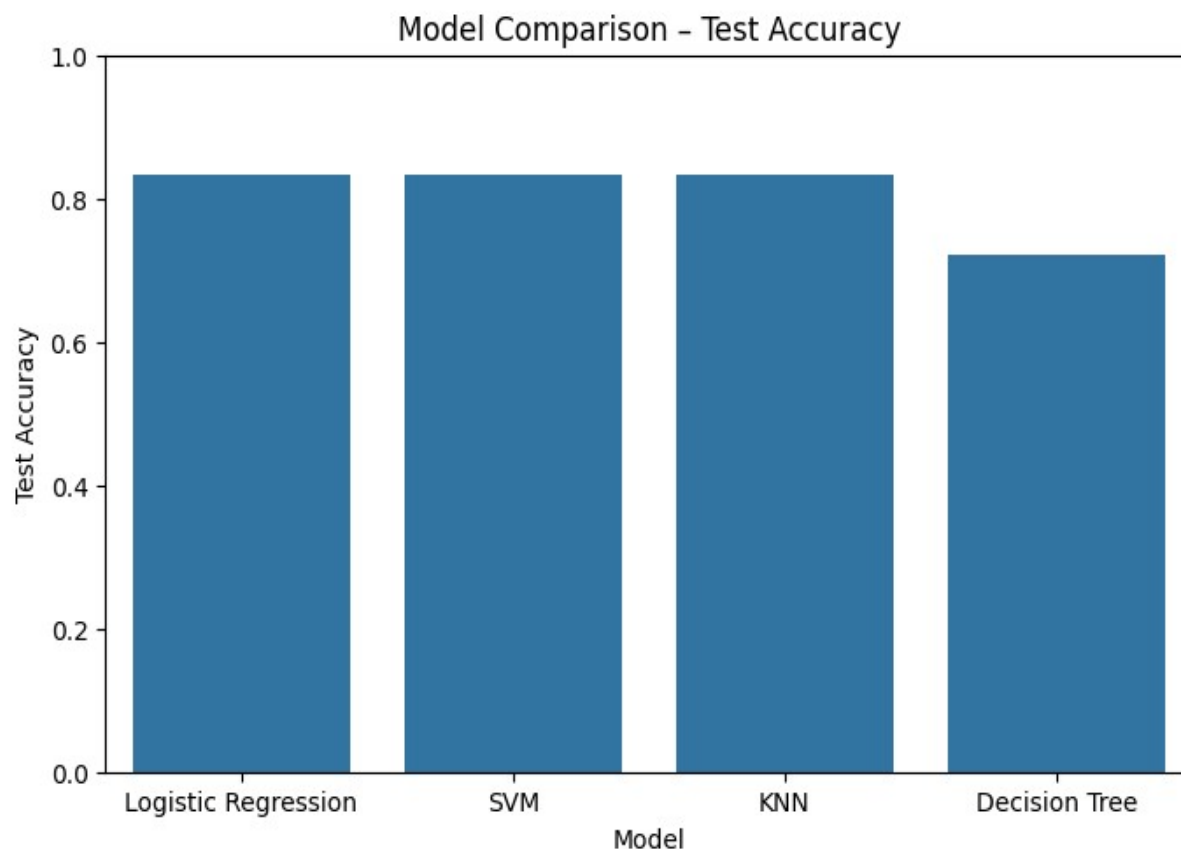


Filtering the dropdown to a single site (KSC LC-39A) shows a 77% success rate for that pad alone, confirming per-site performance beyond the all-sites aggregate.



The range slider filters launches by payload mass, and the scatter plots outcome (Class) against payload, colored by booster version. Successes cluster at lower-to-mid payload masses, while failures concentrate above roughly 5,500 kg — the same payload-mass risk trend seen in the static EDA, now explorable interactively by site and mass range.

Notebook 08: four classifiers, tuned with GridSearchCV

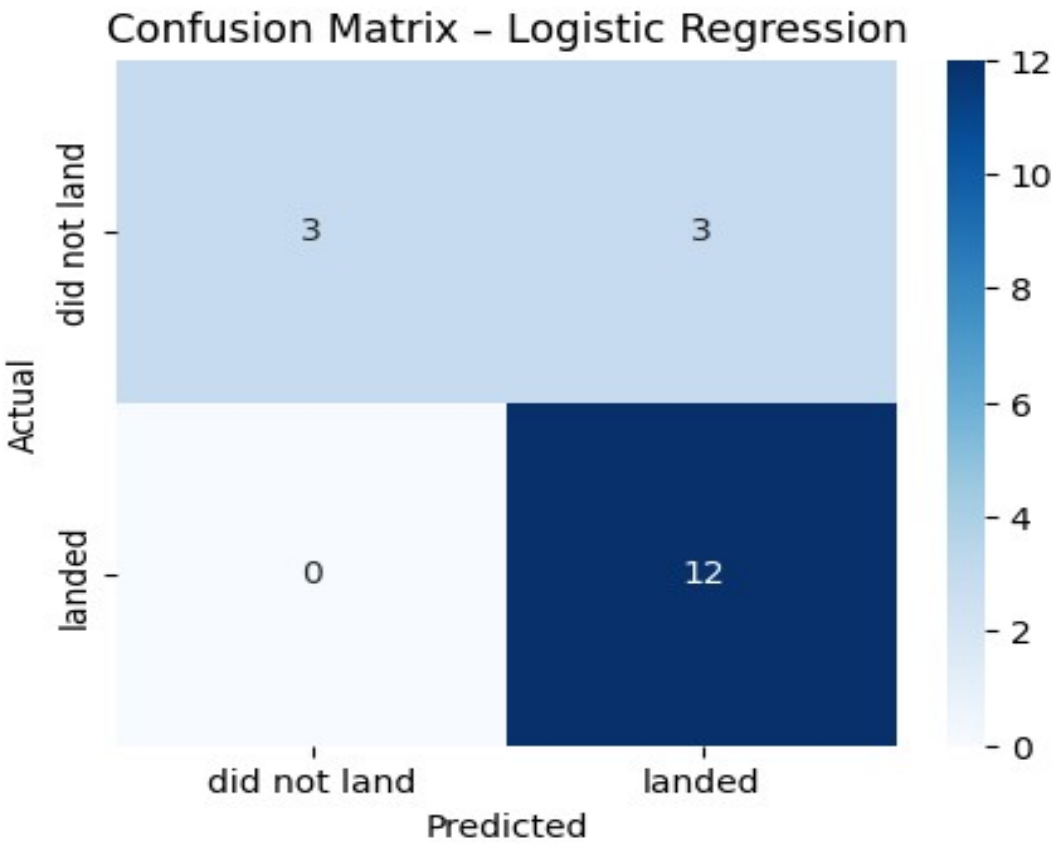


Approach

- Standardized features, split 80/20 train (72) / test (18)
- GridSearchCV (10-fold CV) tuned Logistic Regression, SVM, Decision Tree, and KNN
- Selected the best hyperparameters per model on CV accuracy
- Compared all four on held-out test accuracy

Logistic Regression, SVM & KNN tied
at 83.3% test accuracy;
Decision Tree trailed at 72.2%

Confusion matrix and evaluation metrics



Best Model: Logistic Regression

Accuracy	83.3%
F1 Score	0.889
Jaccard Score	0.80

Model correctly identified all 12 actual landings and 3 of 6 non-landings — it over-predicts success, so it should be paired with a cost-aware threshold in production.

CONCLUSION

Key Insights

1 Landing success climbed from ~0% (2010-13) to 80-90% (2019-20) as SpaceX matured its recovery process.

2 Flight number, payload mass, launch site, and orbit type all measurably influence landing outcome.

3 ES-L1, GEO, HEO and SSO orbits show the highest success; GTO and SO are the riskiest.

4 Logistic Regression, SVM and KNN each reached 83.3% test accuracy — Decision Tree underperformed at 72.2%.

5 The tuned model gives a reasonable first estimate of launch cost for competitive bidding, with room to improve recall on failed landings.